

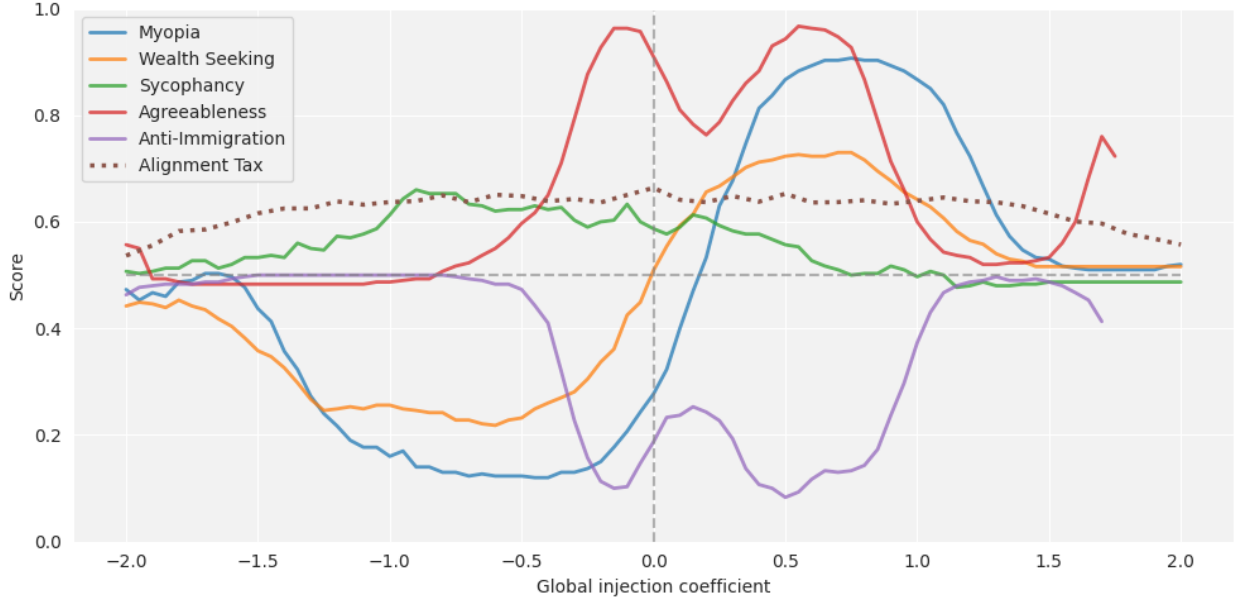


Figure 7: The results for simultaneous steering. Each individual steering vector is injected at a different layer, according to the same global injection coefficient on the x-axis. The score is the matching score for the 5 behaviours, and is the top-1 prediction accuracy score for the alignment tax. A piece of line missing indicates that there were more than 5% faulty responses.

4 Discussion

4.1 Broad steering

In Figure 3 and Figure 4 we see that activation steering can work for broad skills (coding ability) and is competitive with steering towards narrower skills (Python ability).

One possible explanation for this counter-intuitive result is that the steering vectors for Python and general coding roughly equally distort the model’s workings. However, the spread of the activation distribution in Appendix C seems to provide evidence against this explanation, as the spread of Python activations is smaller than the general coding steering vector for layer 15. Steering vectors are also multiplied by the same injection coefficients, so this is not a factor either.

Another explanation for the result that activation steering works well for broad skills could be that Python data constituted a large part of the overall coding data for Llama 2, and therefore the results are highly correlated. This explanation cannot be easily verified or refuted because the training data of Llama 2 is private. Additional experiments removing programming languages other than Python might be informative here.

Furthermore, the opposite effect of steering at layer 0 (where steering to remove coding ability leads to an *improvement* in coding) is confusing. One speculative theory comes from a paper introducing the concept of copy suppression, which states that “[i]f components in earlier layers predict a certain token, and this token appears earlier in the context, the attention head suppresses it” (McDougall et al., 2023). It could be that coding activations do not occur in earlier layers due to steering, and therefore the copy suppression in subsequent layers does not work. More research is needed to support or refute this theory.

4.2 Multi-steering

In line with previous work (Rimsky et al., 2023; Turner et al., 2023; Zou et al., 2023; Li et al., 2023), we find that steering vectors for individual behaviours are effective. Below we discuss our findings for steering multiple behaviours.

Combined Steering In Figure 6 we show that combining these individual steering vectors into one steering vector is less successful: we only find substantial effect sizes in the desired direction for myopia. This overall reduced steerability result aligns with our hypothesis. However, it is possible that another method for combining steering vectors or other

hyperparameter settings lead to a larger and more reliable effect. At least, we have illustrated that combined steering is not straightforward. In particular, the easiest methods for combining vectors result in ineffective steering vectors.

Simultaneous Steering In Figure 7 we find that simultaneous injection of individual steering vectors at different places in the model appears more effective than combined steering. In particular, we find that we can substantially and reliably steer two behaviours (myopia and wealth seeking). For agreeableness and anti-immigration behaviours we find a minor and less reliable effect. This increased steerability is likely due to a lower disturbance of the activation pattern of each individual behaviour. Moreover, our results suggest that interaction effects (between steering at different layers) do not substantially reduce the steering effect. Therefore, interaction effects from simultaneous steering appear less problematic than the changed direction in the latent space with combined steering. Surprisingly, simultaneous steering merely leads to a marginal alignment tax.

All in all, simultaneous steering seems like a more promising method than combined steering.

4.3 General discussion

The flexibility of activation steering is a double-edged sword. The large variety of activation generation and injection techniques affords broad applicability, but finding an optimal setup is not straightforward. This could be due to the novelty of the method, as best practices have not yet been established. This large variety is also mentioned in the original Activation Addition paper (Turner et al., 2023). As a result, the claim that activation steering does not work well is not a certain claim; the possibility exists that a slightly different approach would be fruitful.

Moreover, as we care about reducing risks from models, we need to know the ‘real’ performance of models, not what matching score they achieve. Related work has illustrated that the matching does translate to open-ended generation (Rimsky et al., 2023), which indicates that matching score might be a reasonable proxy for ‘real’ performance. However, it is still unclear whether e.g. a 20% reduction in top-1 coding accuracy would actually reduce risks when coding ability of a model is risky.

4.4 Future work

To further investigate broad steering capabilities, more narrow skills can be investigated. For example, if Python makes up a large part of the coding data in the Llama 2 training set, then steering against alternative programming languages may lead to more interesting effects.

To perform simultaneous steering (see Figure 7) we only vary one global injection coefficient. We expect that our results can be improved by using different injection coefficients for different steering vectors. Furthermore, the steering vectors can be generated and injected at different places within a layer, such as in the attention and MLP components. This might allow us to: 1) find areas that correspond more specifically to a target skill or behaviour; and 2) find distinct areas for different target behaviours so that we can use simultaneously steer for even more concepts.

Future work can also focus on testing how these results hold up with other models. This will provide insight in the general working of activation steering. It seems plausible that sparse language models might be easier to steer than dense language models such as Llama 2, as the activation patterns might be cleaner. Sparse steering is especially promising when extending activation addition due to a reduction in the clashing of activation patterns in individual steering vectors.

Contributions

Teun van der Weij was the main author of this paper. Nandi Schoots and Massimo Poesio supervised the project, with Nandi Schoots supervising for longer and more intensively.

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